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The multi-dimensional performance intelligent evaluation and feedback system for music education

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Abstract

This study seeks to address persistent challenges in traditional music education, particularly the subjectivity inherent in assessment criteria and the frequent delays in feedback, by harnessing the capabilities of artificial intelligence (AI) in conjunction with mobile internet technologies. The primary objective is to design and implement an intelligent, standardized system capable of evaluating musical performances with greater precision and efficiency. To achieve this, the study leverages the WeChat public platform as a foundational interface, upon which a comprehensive, multidimensional intelligent evaluation framework is constructed. This system incorporates advanced deep learning algorithms to facilitate automated and accurate analysis of various instrumental performance techniques. A robust system architecture has been developed, comprising several integrated modules, including user management, a curated digital music library, and intelligent evaluation components. At the algorithmic core of the system lies a neural network-based method tailored for extracting audio features and assessing performance quality, thereby enabling a seamless closed-loop workflow that spans from real-time data acquisition to the generation of intuitive, visualized feedback for learners. Experimental validation confirms that this intelligent evaluation system significantly surpasses conventional methods in terms of both note-level detection accuracy and consistency in full-piece performance scoring. Moreover, the system's auto-play functionality has demonstrated an accuracy rate exceeding 60%, effectively mitigating the prevalent issues of evaluator inconsistency and delayed response in traditional assessment approaches. The key innovation of this study lies in its dual contribution: From a technological standpoint, it integrates professional-grade music performance analysis into a mobile-accessible format. From an educational perspective, it offers objective, prompt, and actionable feedback to learners. Overall, the system exemplifies a forward-thinking application of AI in the field of music education, enhancing instructional quality and advancing fairness, personalization, and adaptability within modern pedagogical contexts. Its implementation carries substantial practical significance for improving both the accessibility and equity of music education in diverse learning environments.



Keywords Music education, Dimensionality reduction, Intelligent assessment, Feedback system, Performance rating

1 Introduction

In the current era of rapid digitalization and intelligent technology development, music education is undergoing a critical transformation from traditional teaching models toward modernized and personalized approaches. Conventional music instruction has largely relied on teachers' subjective evaluations, which, while experiential and flexible, often suffer from inconsistent assessment standards, delayed feedback, and limited quantifiability. These issues are particularly problematic for beginners, who may develop improper techniques in the absence of timely and objective performance feedback, ultimately hindering both learning efficiency and interest. Meanwhile, the widespread adoption of mobile internet and artificial intelligence (AI) technologies has led to a growing demand for intelligent and user-friendly music learning tools. This trend presents significant opportunities for innovation in the field of music education [1].

In recent years, AI technologies have achieved significant progress in areas such as audio signal processing and pattern recognition, laying a solid foundation for the automated evaluation of musical performances [2]. By applying machine learning algorithms, such systems are capable of accurately analyzing multiple performance dimensions, including pitch, rhythm, dynamics, and timbre, and generating objective, quantifiable assessment results [3]. Furthermore, the widespread adoption of mobile internet platforms, such as WeChat public accounts, has greatly facilitated the dissemination of music education tools, enabling learners to upload performance recordings and receive professional feedback anytime and anywhere. However, existing music evaluation systems still exhibit several limitations, including audio quality constraints caused by platform-level compression algorithms, limited evaluation dimensions, and suboptimal user interaction experiences [4]. Therefore, designing a performance evaluation and feedback system based on multidimensional intelligent analysis, while balancing accuracy and usability, holds substantial theoretical significance and practical value.

This study aims to design and implement a multidimensional intelligent performance evaluation and feedback system tailored for music education, addressing issues such as subjectivity in assessment and delayed feedback commonly found in traditional music instruction. The system uses WeChat as the interactive platform and integrates functionalities such as user management through a modular architecture, offering learners a one-stop solution for music learning support. On the technical level, the system employs high-fidelity audio capture to avoid compression loss inherent in native WeChat recordings. It also incorporates signal processing and machine learning algorithms to achieve quantitative analysis across multiple dimensions of performance. The design and implementation of this system not only provide new perspectives for the intelligent transformation of music education but also expand the practical applications of AI in the field of arts education. The main innovations of this study lie in the following two aspects:

First, this study innovatively integrates the correlation features between audio signals and performance actions. It adopts an improved attention mechanism-gated recurrent convolutional neural network (A-GRCNN) to construct a multi-dimensional dynamic evaluation model. A dynamic weight allocation algorithm is used to achieve the

collaborative quantitative analysis of intonation, rhythm, dynamics, and expressiveness. This effectively corrects environmental noise interference. The note-level F-measure reaches 0.94, which is significantly higher than the average level of 0.88 in similar studies. Moreover, it requires no additional hardware support and is compatible with mobile application scenarios.

Second, the study constructs a lightweight intelligent analysis architecture exclusive to the WeChat ecosystem. It innovatively adopts the “mobile phone system recording + web upload” scheme to avoid the native compression defects of WeChat. Combined with adaptive audio compression, model pruning, and quantization technologies, the file size is reduced by 40% and the model size by 65%. Meanwhile, the evaluation time for a single piece of music is controlled within 3 s. Through the closed-loop process of “instruction triggering-asynchronous processing-visual push”, this study achieves convenient interaction without independent application download and real-time accurate evaluation.

2 Literature review

With the rapid advancement of AI and machine learning technologies, the field of music education is undergoing a profound transformation. Current research extends beyond the optimization of unimodal assessment, actively exploring cross-modal integration and alignment with domain-specific perception. These developments provide critical support for achieving greater precision and personalization in music education.

In the construction of evaluation systems, He and Liu proposed a piano performance assessment model based on neural networks. By leveraging machine learning techniques, the system aims to help piano students accurately assess their performance and thereby enhance the overall quality of piano instruction. The model applies signal processing and feature extraction to audio input, evaluating both technical execution and musical expressiveness to generate a comprehensive performance score [5]. Similarly, Liao and Huang developed a music education quality evaluation system utilizing convolutional neural network (CNN). Unlike traditional manual feature extraction methods, CNN is capable of automatically learning deep representations from the input data. Experimental results demonstrated that their model outperformed others in average accuracy on test datasets, highlighting its effectiveness in objectively quantifying teaching quality in music education [6]. Zhang addressed the rise of online music performance in the era of AI. The study emphasized the expressive nature of music, rooted in both composition and performance, and proposed an evaluation system based on the Apriori association rule algorithm. Functional modules were designed to assess online music performances, and experimental results confirmed the feasibility of this approach [7]. Additionally, Chang developed an optimized vocal performance evaluation model incorporating an attention mechanism. By comparing the performance of different models under varying feature inputs, his study revealed that the Attention-Gated Recurrent Convolutional Neural Network (A-GRCNN) achieved the highest scores across F1, accuracy, precision, and recall metrics. The findings suggest that attention mechanisms significantly enhanced the reliability of music information processing and the accuracy of vocal performance evaluation [8]. Liu et al. further addressed the challenge of aligning automated assessment with human perception by introducing a novel evaluation framework based on the TTM model. The study introduced MusicEval, the first generative music assessment dataset, consisting of 2,748 music samples generated by 31 advanced

models from 384 textual prompts, annotated with 13,740 scores from 14 expert musicians. This dataset laid the foundation for more perceptually aligned and scalable music assessment systems [9].

In terms of music evaluation methodologies, Gao introduced a novel multimodal fusion-based assessment approach that integrated auditory information and motion data to address the challenges of evaluating students' learning status in university-level music education. By incorporating Internet of Things (IoT) technologies, the proposed framework offered new insights for future interactive classroom assessments and expanded the application scope of multimodal information fusion in music pedagogy [10]. Subsequently, Cao and Liew conducted a comparative study between traditional assessment techniques and machine learning-based methods in evaluating music education quality. They proposed a machine learning-driven evaluation framework, highlighting that the integration of multiple assessment strategies can lead to more comprehensive and accurate results [11]. Fujii and Samm investigated the influence of musical expertise on perceptual bias through a controlled study based on recordings from Japanese high school brass band competitions. The findings revealed that carefully controlling for visual elements and musical works could significantly reduce the dominance of visual cues in performance evaluation [12]. In another direction, He and Dong introduced a Layered Fuzzy Logic Approach (LFLA) to assess the musical expressiveness of vocal teaching methods. This approach aligned evaluation criteria with specific learning guidelines, which varied across different music curricula and vocal genres. By analyzing instructional sequences through hierarchical fuzzy logic, the method offered refined insights into the actual learning process and facilitated pedagogical adjustments [13]. Furthermore, Plitsis et al. combined quantitative assessments of music-specific features with qualitative user studies. Their analysis demonstrated that similarity metrics often aligned with user preferences and that current personalization methods tended to be more effective in supporting rhythmically driven musical structures than melodic ones [14]. In addition, in the field of intelligent evaluation for music education in recent years, two other important research results are worthy of attention. Deshpande et al. compared the performance of five machine learning models (including Support Vector Machine (SVM) and Random Forest (RF)) in sentiment analysis of teacher feedback. They proposed that complex models (e.g., Random Forest) were more suitable for accurate analysis in educational scenarios. Their research idea provided a reference for the quantitative modeling of multi-dimensional features in music education evaluation, specifically, verifying the effectiveness of core algorithms through multi-model comparison to avoid the limitations of a single algorithm [15]. In another study, Deshpande et al. proposed a context-aware mobile learning content recommendation method based on the Adaptive Neuro-Fuzzy Inference System (ANFIS). By integrating the advantages of fuzzy logic and neural networks, this method effectively processes ambiguous and uncertain context data in educational scenarios (such as learning environment noise and network status). Its cross-modal data fusion idea provided important technical inspiration for solving the problem of "collaborative analysis of multi-dimensional features (pitch/rhythm/dynamics) in music performance", especially with reference significance in handling weight allocation and error compensation across different feature dimensions [16].

In summary, recent years have witnessed the emergence of various intelligent evaluation methods in the field of music education. Leveraging AI technologies such as neural

networks and attention mechanisms, significant progress has been made in areas such as performance technique analysis and instructional effectiveness assessment. However, existing systems still face limitations, including narrow evaluation dimensions and insufficient multimodal integration. This study proposes an innovative cross-modal dynamic evaluation system that integrates multi-source data to significantly enhance the comprehensiveness and personalization of music performance assessment. The proposed approach offers a more precise and holistic automated evaluation solution for music education, with strong practical implications for improving teaching quality and promoting personalized learning.

3 Design of multi-dimensional performance intelligent evaluation and feedback system for music education

3.1 System functional requirements analysis

3.1.1 Overall requirement

The functional requirements analysis for a multidimensional intelligent performance evaluation and feedback system tailored to music education mainly focuses on user interaction, resource management, and intelligent assessment [17]. The system should support users in setting up personal profiles and accessing music library resources, including scores and reference audio, thereby providing learners with standardized learning materials. Core functionalities require users to upload performance audio, which the system analyzes through intelligent algorithms across multiple dimensions to generate scores and provide improvement suggestions. The overall design is based on the WeChat platform to enable convenient mobile operation, meeting the needs for real-time evaluation and personalized guidance in music education. The detailed system requirements are illustrated in Fig. 1.

3.1.2 Detailed requirements analysis

In the overall system requirements diagram in Fig. 1, the multidimensional intelligent performance evaluation and feedback system for music education encompasses seven key functional requirement dimensions. These requirements collectively form the

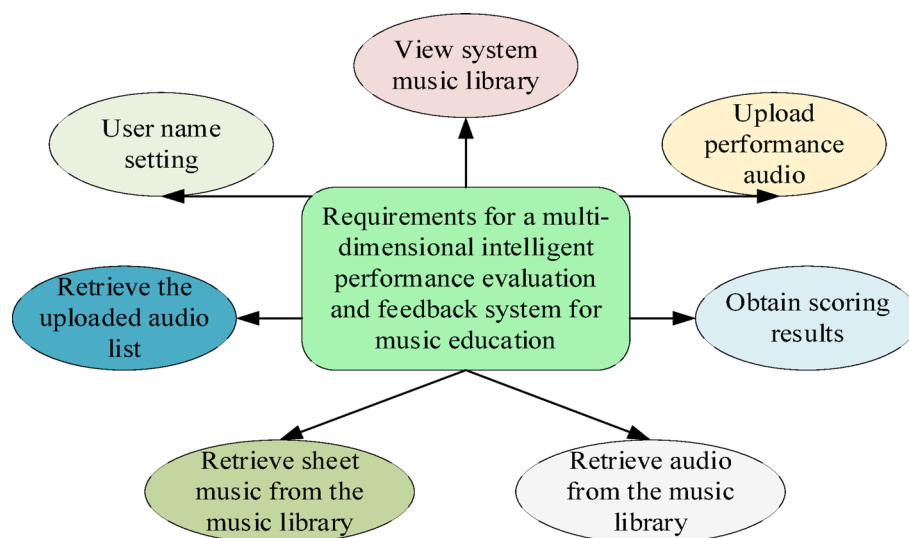


Fig. 1 Schematic diagram of overall system requirement

system's fundamental architecture and core functional modules, reflecting the complete user interaction process and service demands. Each requirement dimension is analyzed in conjunction with specific use cases and technical implementation strategies to ensure comprehensiveness and feasibility. This structured approach to requirements analysis not only clarifies the detailed specifications of each functional module but also provides clear guidance for subsequent system design and development. Details are presented in Table 1.

Table 1 summarizes the main functional modules and interaction processes of the music performance scoring system based on the WeChat Official Account platform. From the perspective of user identity management, the system utilizes the WeChat OpenID as a unique identifier and enhances interaction friendliness by guiding users to set personalized usernames. Regarding core functionalities, the system establishes a comprehensive music resource service framework: users can obtain information about the system's music library and download standardized scores and professional performance audio via standardized message commands. These foundational resources provide necessary references for subsequent performance scoring. Notably, to ensure scoring accuracy during audio collection, the system innovatively avoids the AMR compression limitation of WeChat's native recording interface. Instead, it adopts a hybrid solution combining phone system recording and web uploading. This design choice reflects the system's strict requirements for audio quality. In terms of data management, the system implements complete tracking of user upload records, distinguishing between scored and pending scoring statuses. Professional scoring results are presented through visualized charts, significantly enhancing result readability. The entire system's interaction design is implemented based on the WeChat message framework, aligning with users' existing habits while ensuring data security and privacy through a strict OpenID binding mechanism. This architecture leverages the dissemination advantages of the WeChat platform while meeting the precise evaluation needs of music learners through specialized functional design.

3.2 System function module design

The system is built on the WeChat Official Account platform, where users interact by sending specific commands. The server responds with text, images, or audio based on the request type. The system adopts a modular design comprising three main functional

Table 1 Introduction of each function

Functional module	Mode of operation	System response
User name setting	After paying attention to WeChat official account, set the user's name as prompted.	The user's name is bound to WeChat openid as the user identity.
View system music library	Send a message in the specified format to WeChat official account.	Return to the list of songs supported by the system.
Get the score of the music library	Send a message containing the names of the songs in the library.	Returns the score picture of the corresponding music.
Get music library audio	Send a message marked "Audio" request.	Return the standard performance audio in the form of WeChat voice
Upload playing audio	Upload local recording files of mobile phones through WeChat official account page.	The server receives the file and stores it in the user record.
Get the upload audio list	Send a designated request message to WeChat official account.	Returns a list of all uploaded audio of the user.
Get the scoring results	Send a message containing the scored audio file name.	Return the graphical scoring results in the form of pictures.

modules: WeChat interaction module includes automatic reply and menu navigation functions, responsible for processing user commands. Audio preprocessing module covers message communication, format conversion, and music recognition to prepare audio data. Performance scoring module involves segment localization, multi-track detection, note evaluation, and report generation to achieve professional scoring analysis. These modules collaborate seamlessly, ultimately delivering the scoring results to users as visualized images. The system architecture is illustrated in Fig. 2.

3.2.1 Network architecture design and basic layer configuration

The core evaluation model of the system adopts a hybrid network architecture of “CNN + BiLSTM + multi-task learning”. The specific configuration is as follows: The CNN part consists of 3 convolutional layers: The first layer is equipped with 32 convolution kernels of size 3×3 , with a stride of 1 and padding mode set to “same”, using ReLU as the activation function. The second layer is upgraded to 64 convolution kernels of size 3×3 , maintaining the same stride and padding mode, and also using ReLU activation. The third layer adopts 128 convolution kernels of size 3×3 , followed by a Batch Normalization layer and a Dropout layer (dropout rate = 0.3) to prevent overfitting. After the CNN layers, Global Average Pooling is used to convert feature maps into 1D feature vectors with a dimension of 128. The BiLSTM part comprises 2 bidirectional long short-term memory network layers: Each layer has 128 hidden units. Inter-layer processing includes Dropout (dropout rate = 0.2) and Layer Normalization. The input sequence length is fixed at 512 (corresponding to the number of audio frames), and the output dimension is 256 (bidirectional feature concatenation). The multi-task output layer includes two branches: Main task branch: Fully connected layer (128 neurons, ReLU activation) + regression layer (outputs a single comprehensive score ranging from 0 to

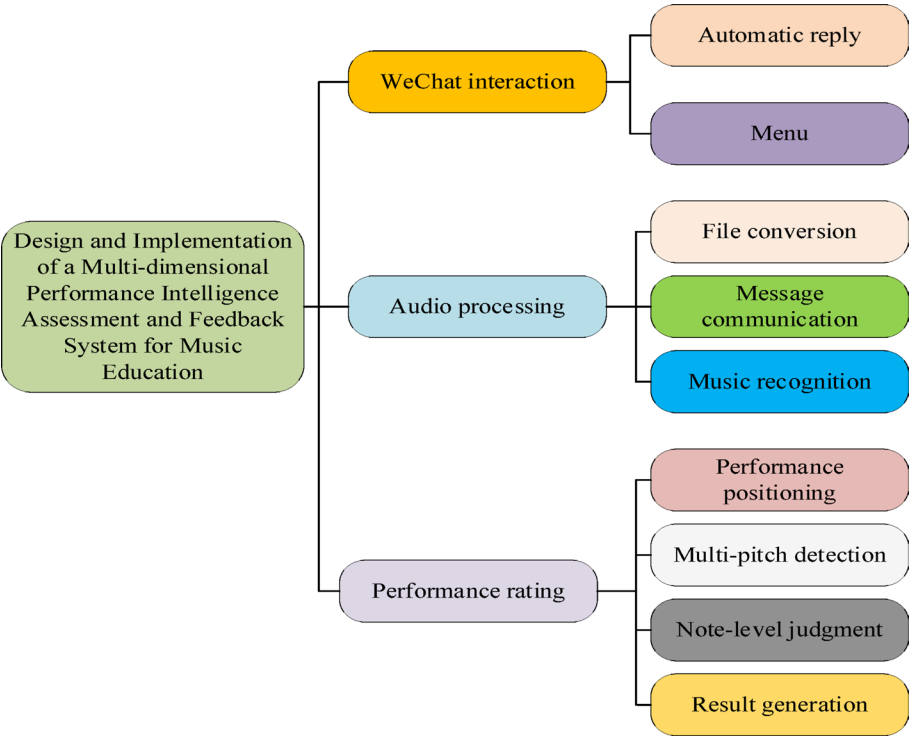


Fig. 2 Schematic diagram of system function module design

100), using Mean Squared Error (MSE) as the loss function. Auxiliary task branch: Fully connected layer (64 neurons, ReLU activation) + softmax classification layer (outputs 4 error types: intonation error, rhythm error, dynamics error, timbre error), using Cross Entropy as the loss function. The loss weight ratio of the two branches is 1:0.5. The overall parameter scale of the model is approximately 2.8 M: The CNN part accounts for 35% (about 980 K). The BiLSTM part accounts for 52% (about 1.46 M). The output layer and other parameters account for 13% (about 360 K). The model supports lightweight deployment on mobile devices.

3.2.2 Technical details of the WeChat interaction module

The WeChat interaction module is implemented using the WeChat Official Account Development Interface (based on WeChat API v3). It consists of two sub-modules: an auto-reply engine and a custom menu navigation. The auto-reply engine supports keyword matching, regular expression parsing, and event triggering. Its response delay is controlled within 100–300 ms. The custom menu adopts a three-level structure design: The first-level menu includes “My Learning”, “Music Library Resources”, and “Help Center”. The second-level menu refines function entrances. Webpage jumps are implemented via WeChat JS-SDK to ensure cross-device compatibility.

3.2.3 Technical details of the audio preprocessing module

The audio preprocessing module is implemented with Python + FFmpeg, and its core workflow includes message communication, format conversion, and music piece recognition. Message communication: Receives audio files uploaded by users through the callback interface of the WeChat server, adopts the HTTPS protocol to ensure transmission security, and supports resumable uploads. Format conversion: Uniformly converts user-uploaded WAV/MP3 files to the standard PCM format via FFmpeg command-line tools, with a conversion error rate of <0.1%. Music piece recognition: Matches the system music library based on audio fingerprint technology, extracts the spectral features of the audio and compares them with the fingerprint library of reference audio in the music library. The matching threshold is set to 0.85, with recognition time < 3 s per piece and accuracy > 98%.

3.2.4 Technical details of the performance scoring module

The performance scoring module serves as the core of the system, implemented based on a three-level architecture of “feature extraction - multi-dimensional evaluation - report generation”. Paragraph positioning: Adopts the Dynamic Time Warping (DTW) algorithm to align the user’s performance audio with the reference audio, dividing the entire piece into several paragraphs, with a paragraph boundary detection accuracy of > 95%. Multi-track detection: Extracts pitch features based on the improved CREPE model, and combines the librosa library to extract rhythm, dynamics, and timbre features, enabling multi-dimensional parallel detection. Note evaluation: Constructs a binary classification model to score the “correctness” and “completeness” of each note, with the evaluation time per note < 1 ms. Report generation: Uses Matplotlib to draw a multi-dimensional scoring radar chart, combined with a time axis marked with errors, to generate a PNG format report compatible with mobile phone screens.

3.2.5 Correlation logic between functional modules and top-level architecture

The efficient operation of the above functional modules relies on the support of a unified underlying software architecture. As the interaction entry between users and the system, the WeChat interaction module requires interface adaptation for its message processing, menu navigation, and other functions through the WeChat interaction sub-layer in the application layer of the top-level architecture. It also depends on the user service in the service layer to complete identity authentication and permission management. Compute-intensive tasks of the audio preprocessing module (such as format conversion and music piece recognition) need to rely on computing resources (cloud server GPU) in the infrastructure layer for efficient processing. The processed audio data must be securely stored and quickly retrieved through distributed file storage in the data layer. As the core business module, the performance scoring module implements containerized deployment of its core logic (including multi-dimensional feature extraction and intelligent scoring) through the evaluation service in the service layer. This ensures response speed under high-concurrency scenarios. It also relies on the MySQL database in the data layer to store scoring results and user learning data, providing data support for subsequent personalized feedback generation. The functional modules designed in this section clarify the core business boundaries of “what the system does”. The top-level architecture in Sect. 3.3 will define the underlying support system of “how to implement it”. Through the corresponding relationship of “business functions-architecture services”, the two form a close linkage, ensuring the efficient and stable operation of the entire system from user interaction to intelligent evaluation.

3.3 System top-level design

3.3.1 Introduction to the system's top-level software architecture

The system's top-level software architecture adopts a layered design pattern, divided into the Application Layer, Service Layer, Data Layer, and Infrastructure Layer from top to bottom. Each layer has clear responsibilities and interacts through standardized interfaces to ensure system scalability and maintainability.

3.3.2 Application layer

As the user-oriented interaction entry, it is implemented based on the WeChat Official Account ecosystem, including two sub-layers: the WeChat Interaction Sub-layer and the Visual Feedback Sub-layer. The WeChat Interaction Sub-layer provides message instruction interaction, custom menu navigation, and web-based functions, adapting to mobile screen sizes. The Visual Feedback Sub-layer generates multi-dimensional scoring reports and personalized improvement suggestions, supporting report export.

3.3.3 Service layer

As the core business logic processing layer of the system, it adopts a microservice architecture to split modules, including User Service, Resource Service, Evaluation Service, and Notification Service. User Service is responsible for user identity management and learning data statistics, implementing identity authentication using JWT tokens, with data stored in a MySQL database. Resource Service manages music library resources, enabling resource upload, retrieval, and update. It uses Redis to cache popular resources, achieving a cache hit rate of >80%. Evaluation Service performs audio preprocessing,

multi-dimensional feature extraction, and intelligent scoring. It is deployed in Docker containers to support horizontal scaling. Notification Service pushes operation results and learning reminders to users, supporting two push methods: WeChat template messages and official account articles, with a push success rate of > 99%.

3.3.4 Data layer

Responsible for data storage and management, it adopts a multi-database collaboration scheme, including MySQL database, MongoDB database, and distributed file storage. MySQL database stores structured data, using the InnoDB engine to support transactions, with table structure designed in compliance with the Third Normal Form. MongoDB database stores unstructured/semi-structured data, supporting flexible field expansion. Distributed file storage adopts a multi-replica strategy to ensure data reliability and supports file lifecycle management.

3.3.5 Infrastructure configuration

This study provides the underlying support required for system operation, including computing resources, network resources, and security protection. Computing resources: Cloud servers are used, with hardware configurations including Intel Xeon Gold 6248 CPU, NVIDIA T4 GPU, 32GB DDR4 memory, and 1 TB SSD. The operating system is Ubuntu 20.04 LTS, supporting elastic scaling. When the model is deployed on the WeChat platform, it actually runs on the server side, relying on the above cloud server configuration to undertake core computing tasks. The user's mobile terminal is only responsible for interactive operations and data transmission. To ensure a smooth user experience across mobile devices with different performance levels, the system adopts a layered adaptation strategy: For high-performance mobile phones: Optimizes data transmission rate to support real-time upload of high-definition audio. For mid-to-low-performance mobile phones: Automatically adapts to low-bitrate transmission mode, simplifies redundant rendering of the front-end interface, and reduces mobile terminal computing consumption through server-side precomputation and caching mechanisms. This ensures that the total response delay from audio upload to feedback generation is controlled within 1200 ms. Network resources: Configured with 100 Mbps bandwidth, using CDN to accelerate static resources, with global access delay < 500 ms. Security protection: Deploys WAF to defend against SQL injection and XSS attacks, adopts data encryption (AES-256 encryption for user sensitive information, HTTPS encryption for transmitted data), and complies with the requirements of the "Personal Information Protection Law".

According to the top-level block diagram, the system is divided into several modules: storage module, clock division module, address generator module, programmable frequency divider module, frequency division preset module, and music playback module.

Storage module: Stores the note data of the music piece.

Music playback module: Plays the final generated audio.

Frequency division preset module: Provides the initial count value for the programmable frequency divider module.

Address generator module: Acts as the address generator for the note ROM data in the storage module.

Programmable frequency divider module: Returns the frequency corresponding to each note based on the preset division value.

Clock division module: Provides clock pulses to the programmable frequency divider module. It processes a 12 MHz clock pulse through a 16-times division to obtain a frequency of 750 kHz. The block diagram is shown in Fig. 3.

3.4 Introduction of top-level functions of the system

3.4.1 Storage function

The storage function is realized through the collaboration of distributed file storage and MySQL database. Music score storage and management: Supports users to upload custom music scores. After format verification, the scores are stored in distributed file storage, and metadata is written into the “Music Score Table” of MySQL. Users can retrieve and download scores through the “My Music Scores” function. Performance data comparison: Automatically associates the audio uploaded by users with the corresponding reference audio. Feature data of both is stored in the “Performance Comparison Table” of MongoDB, enabling users to view feature difference graphs. Environmental protection and convenience: Replaces paper music scores with digital ones, supporting page-turning via gesture control to reduce paper consumption. The system regularly pushes “environmental protection reports” to users (Fig. 4).

3.4.2 Automatic playback function

The automatic performance function serves as the system’s “reference audio generation tool”, implemented based on MIDI synthesis technology. Its core logic consists of three components: a main control module, a MIDI synthesis module, and an audio output module. Main control module: Written in C language, it includes two sub-modules: The main logic control sub-module is responsible for functional priority scheduling. The playback logic control sub-module reads MIDI data of the corresponding music piece and parses it into control instructions. MIDI synthesis module: Implements MIDI message generation using the open-source PortMidi library, supporting the GM standard tone library. It allows tone switching according to user needs and controls the dynamic

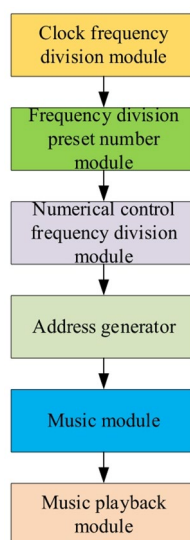


Fig. 3 Schematic diagram of top-level design of the system

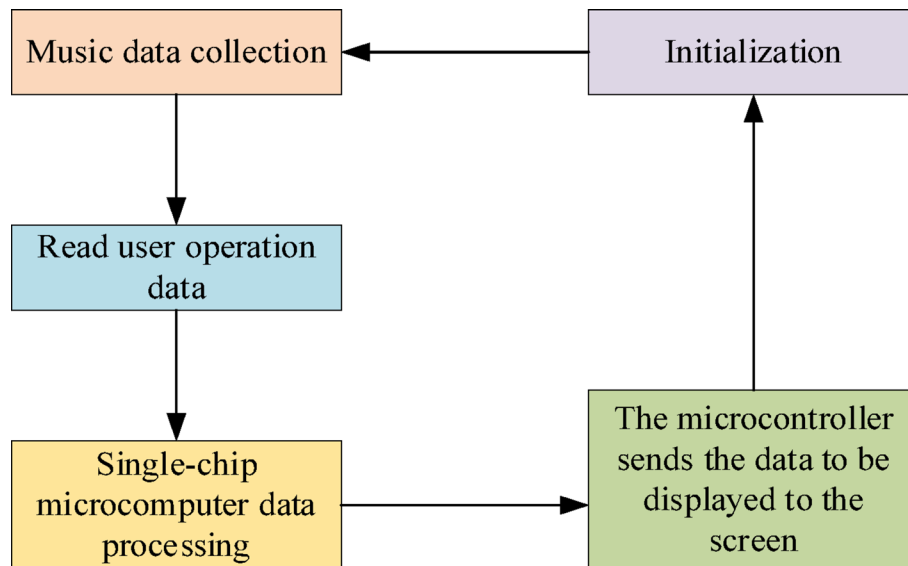


Fig. 4 Schematic diagram of storage function operation framework

variation of tones via ADSR envelope. Functional implementation: Automatic playback, music piece switching, and pause functions are realized through “timestamp control”. Playback delay is controlled within 10–30 ms to ensure rhythmic stability.

The three sub-functions, automatic playback, track switching, and pause, are implemented together. Automatic playback and stop functions are controlled via timing delays. The flowchart of the automatic playback program is shown in Fig. 5.

3.4.3 Automatic evaluation function

The automatic evaluation function serves as the core of the system, implemented based on a closed-loop process of “data collection - feature extraction - multi-dimensional scoring - feedback generation”. Data collection phase: Users record performance audio via the mobile phone’s system recording function to avoid compression losses from WeChat’s native recording. After recording, the audio is uploaded to the server through the official account’s web interface, supporting resumable uploads. Preprocessing phase: The server performs format verification after receiving the audio, converts it to PCM format via FFmpeg, and finally uses a Butterworth low-pass filter to remove high-frequency noise. The signal-to-noise ratio (SNR) is improved by ≥ 10 dB. Feature extraction phase: Multi-algorithm collaboration is adopted to extract multi-dimensional features, including pitch features, rhythm features, dynamics features, and timbre features. Scoring phase: A multi-dimensional scoring model is constructed. Weights of each dimension are determined by the Analytic Hierarchy Process (AHP), and the final score is presented on a 100-point scale. Feedback generation phase: The system automatically generates visual reports and personalized improvement suggestions, which are pushed to users via WeChat messages. The total processing time is < 60 s.

Combined with the above-mentioned system operation process, the process of multi-dimensional performance intelligent evaluation and feedback system for music education mainly has five steps, and the process of the system is shown in Fig. 6.

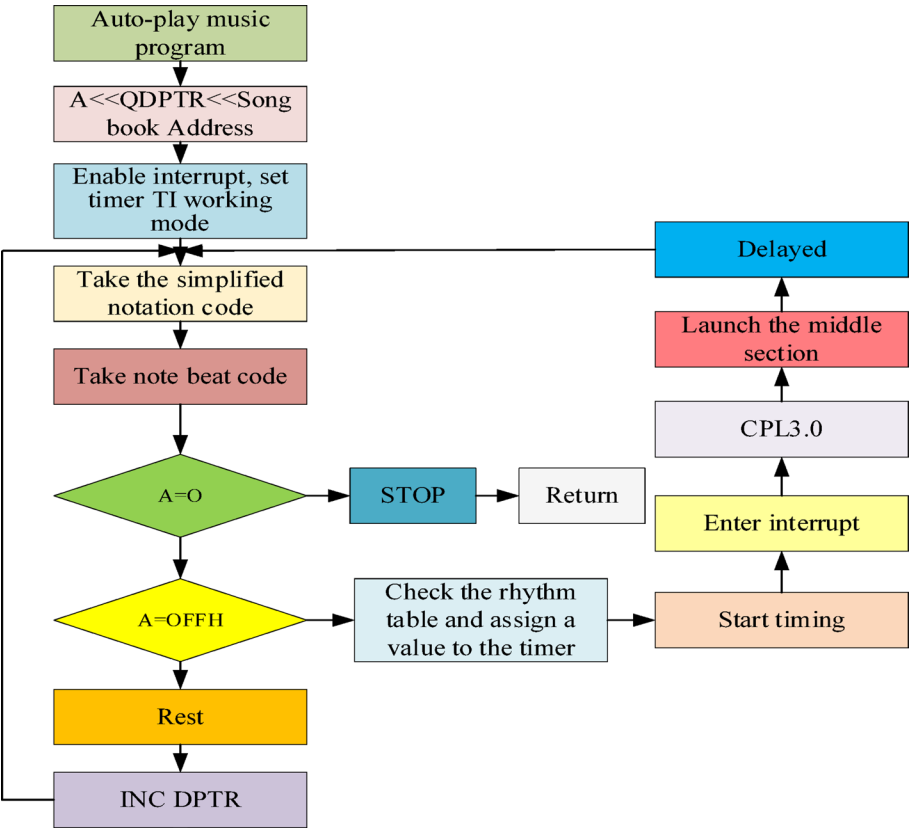


Fig. 5 Schematic diagram of automatic playback process

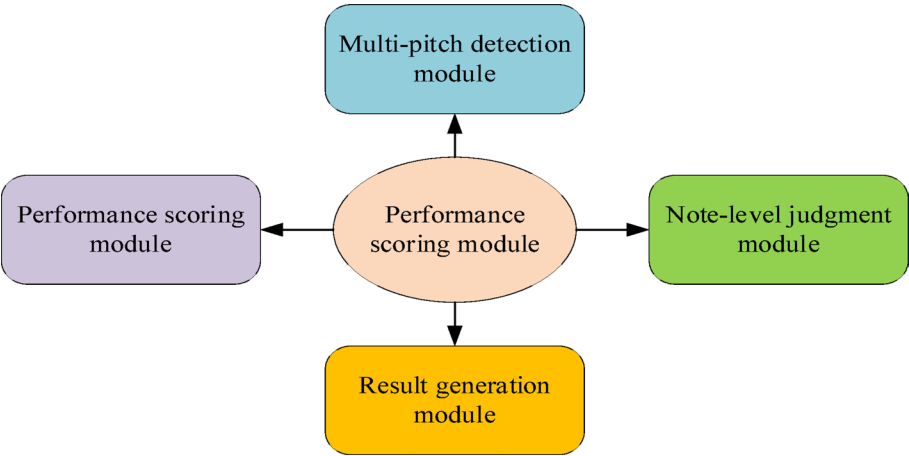


Fig. 6 Schematic diagram of performance scoring module

4 The realization and test of multi-dimensional performance intelligent evaluation and feedback system for music education

4.1 System implementation flow

The system implementation process consists of four core stages: data acquisition, feature extraction, intelligent evaluation, and feedback generation (Fig. 7). First, multi-modal data from the performer, such as audio signals, fingering movements, and force dynamics, are collected using high-precision sensors and audio devices, ensuring both

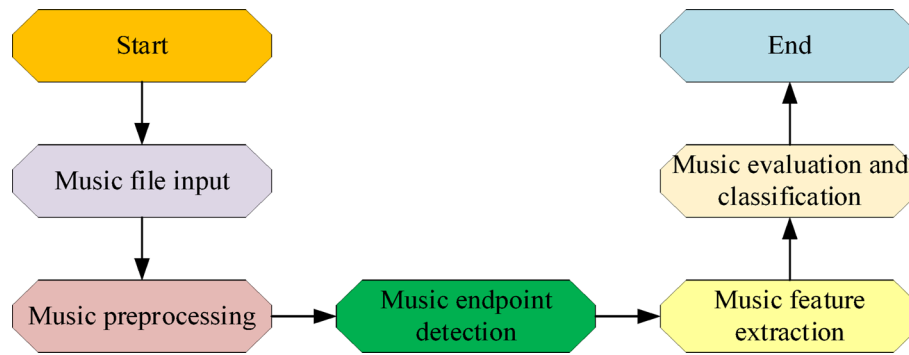


Fig. 7 Flow diagram of automatic evaluation system

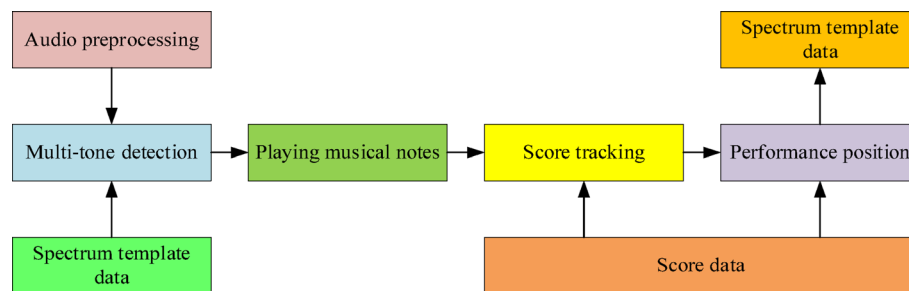


Fig. 8 System implementation flow

comprehensiveness and accuracy of the input. Second, signal processing techniques and machine learning algorithms are employed to extract relevant features from the raw data, including musical attributes such as pitch, rhythm, and timbre, as well as motor features like fluency and coordination of hand movements, thereby providing a quantitative foundation for subsequent evaluation. Next, based on the extracted features, a pretrained deep learning model performs multidimensional intelligent evaluation, covering technical accuracy, musical expressiveness, and emotional delivery, ultimately generating a detailed scoring and analysis report. Finally, the system automatically produces personalized feedback based on the evaluation results. They include visualized error highlights, targeted improvement suggestions, and customized practice recommendations, thereby helping performers enhance their performance with precision. This end-to-end closed-loop process provides efficient and scientific intelligent support for music education (Fig. 8).

4.1.1 Data collection process

In addition to audio signals, the multi-modal data collection adds a “performance action data” collection function. Users invoke the mobile phone camera via the WeChat Mini Program to record videos of hand movements during performance. The system extracts hand key points through OpenCV to analyze fingering fluency. Audio signal collection adopts mobile phone system recording to avoid compression losses of WeChat’s native AMR format, ensuring audio quality. Data annotation and quality control are performed manually by 3 music professionals. A Kappa coefficient >0.85 is deemed consistent annotation. Annotation content includes intonation error positions, rhythm delay time, and fingering error types. The system automatically filters low-quality data to ensure the

quality of training data. Dataset construction: A “multi-dimensional music performance dataset” is built, containing 50 music pieces covering styles such as classical, pop, and jazz, with difficulty levels ranging from 1 to 10. Each piece includes 100 user performance samples, totaling 5,000 samples. Sample attributes include performer level, performance equipment, and recording environment, providing support for verifying the model’s generalization ability.

4.1.2 Introduction to key information of the dataset and data collection

The dataset constructed in this study consists of 100 performance samples, covering 50 music pieces of different styles. Ten representative pieces are selected for testing in the experimental section. Recording conditions: Audio recording uses the native recording function of mobile phone systems. The recording environment must be free from obvious sudden noise interference: noise ≤ 30 dB in quiet environments, and noise controlled between 60 and 70 dB in noisy environments. Performers are required to play completely according to standard scores, avoiding intentional pauses or adaptations to ensure sample validity. Audio parameters: The unified sampling format is 44.1 kHz sampling rate, 16-bit depth, and mono channel. Supported file formats include WAV and MP3, which are converted to PCM format through preprocessing for subsequent feature extraction. Ground truth label acquisition: Adopts a “manual annotation + cross-validation” mode. Three music professionals with teaching experience ≥ 5 years are invited to independently annotate samples in dimensions such as intonation correctness, rhythm accuracy, and dynamics matching. Annotation content includes error positions, error types, and correction suggestions.

4.1.3 Feature extraction stage

Audio feature extraction includes pitch features, rhythm features, dynamics features, and timbre features. Pitch features: The CREPE model is used to extract pitch sequences, which are converted to cents to calculate deviations from reference pitches. Rhythm features: The librosa.beat.beat_track() function is used to calculate BPM. The Dynamic Time Warping (DTW) algorithm is then adopted to align the rhythm sequences of the user’s performance and the reference audio, and the time deviation of each beat is calculated. Dynamics features: Short-time energy is extracted and converted to dB values to calculate deviations from reference dynamics. The equation for short-time energy is:

$$E_n = \sum_{i=1}^N x_{n,i}^2 \quad (1)$$

$x_{n,i}^2$ is the amplitude of the i -th sampling point of the n th frame, and N is the frame length. The conversion equation of dB value is

$$dB = 20\log_{10}(E_n/E_0) \quad (2)$$

E_0 represents the reference energy. For timbre features, 13-dimensional MFCCs are extracted. First-order and second-order differences of the MFCCs are calculated to construct a 39-dimensional timbre feature vector. Principal Component Analysis (PCA) is adopted for dimensionality reduction to 20 dimensions, reducing computational complexity. For motion feature extraction, the MediaPipe Hands library is used to extract

coordinates of 21 key points of the hand. Finger joint angles and finger movement speeds are calculated to evaluate fingering fluency.

4.1.4 Intelligent evaluation stage

The model architecture adopts a deep learning model of “feature fusion + multi-task learning”, with the basic framework of “CNN + BiLSTM”. CNN consists of 3 convolutional layers with 32/64/128 filters and a kernel size of 3×3 , used to extract local features. BiLSTM comprises 2 layers with 128 hidden units, used to capture temporal dependencies. The output layer adopts a multi-task design: the main task is “comprehensive score prediction”, and the auxiliary task is “error classification across dimensions”. Model training: Uses the “multi-dimensional music performance dataset”, split into training and test sets at an 8:2 ratio. Data augmentation is applied to the training set to avoid overfitting. Implemented with the PyTorch framework, using the Adam optimizer (learning rate = 0.001, weight decay = $1e-5$). The loss function is a combination of “MSE loss + Cross Entropy loss” with a weight ratio of 1:0.5. Batch size is set to 32, training epochs to 50, with an early stopping strategy adopted. Post-training: The model is converted into a serialized file via TorchScript and deployed to the server side. TensorRT is used to accelerate inference. Evaluation metrics: In addition to the traditional note-level F-measure, dimension-specific evaluation metrics are added, including intonation accuracy, rhythm accuracy, dynamics matching degree, and timbre similarity.

4.1.5 Core parameters of model training and data processing strategies

The “CNN + BiLSTM + multi-task learning” evaluation model constructed in this study is trained on a multi-dimensional music performance dataset containing 50 pieces of music with different styles and difficulty levels, totaling 5,000 performance samples. The dataset is split into a training set (80%, 4,000 samples) and a test set (20%, 1,000 samples). Samples cover styles including classical, pop, and jazz, with difficulty levels ranging from 1 to 10. Each piece corresponds to 100 audio samples and hand movement video data from performers of varying skill levels. Data preprocessing phase: Audio data is first converted to PCM format with a 44.1 kHz sampling rate and 16-bit depth via FFmpeg, then a Butterworth low-pass filter is applied to remove high-frequency noise, improving the signal-to-noise ratio (SNR) by ≥ 10 dB. Motion data is processed by extracting coordinates of 21 hand key points using the MediaPipe Hands library, which are normalized to the 0–1 range to construct feature vectors. Model training configuration: Batch size is set to 32, with an initial learning rate of 0.001. The Adam optimizer (weight decay coefficient = $1e-5$) is used for parameter updates. The loss function adopts a multi-task joint design: mean squared error (MSE) loss for the main task (comprehensive score prediction), and cross-entropy loss for the auxiliary task (error type classification), with a weight ratio of 1:0.5 between the two. Training is set to 50 epochs, with an early stopping strategy (training terminates if validation set loss does not decrease for 5 consecutive epochs) to avoid overfitting. Data augmentation strategies: For audio samples: random tempo adjustment ($\pm 10\%$ BPM), random transposition (± 2 semitones), and addition of low-intensity background noise (SNR = 30–40 dB). For motion samples: random cropping and minor perturbations of key point coordinates. These strategies effectively enhance the model’s generalization ability.

4.2 Result test

4.2.1 System application testing results

Given that the multidimensional intelligent performance assessment and feedback system for music education is primarily designed for piano learners and aims to support their musical performance, the system testing process places particular emphasis on effectiveness, practicality, and the performance of the automated assessment algorithm when applied to actual music data. The testing methodology is developed based on system requirements, with predefined expectations established in advance. The results of the system application tests are presented in Table 2.

4.2.2 Note accuracy test results

One of the commonly used evaluation metrics for assessing the accuracy of multi-pitch detection is the note-level F-measure. A higher F-measure value indicates greater detection accuracy. The calculation methods are presented in Eqs. (3), (4), and (5).

$$\text{Recall} = \frac{n_{\text{Corr}}}{n_{\text{Ref}}} \quad (3)$$

$$\text{Precision} = \frac{n_{\text{Corr}}}{n_{\text{Tot}}} \quad (4)$$

$$\text{F measure} = \frac{2\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (5)$$

n_{Corr} denotes the total number of correctly detected notes, n_{Ref} represents the total number of notes in the reference (ground truth) audio, and n_{Tot} indicates the total number of detected notes. Figure 9 illustrates the note-level evaluation results for the test samples.

In Fig. 9, the note-level evaluation data of 10 music pieces covers three difficulty levels (low, medium, and high). The number of reference notes ranges from 95 to 147, including 4 low-difficulty pieces, 4 medium-difficulty pieces, and 2 high-difficulty pieces. The overall data presents a reasonable distribution: the number of correctly detected notes is slightly lower than the number of reference notes, and the total number of detected notes is slightly higher than the number of reference notes. The average difference between correctly detected notes and reference notes is 4.8, with a matching degree of 95.7%. For high-difficulty pieces: Piece 5 has 145 correctly detected notes and 147 reference notes, achieving a matching degree of 98.6%. Piece 8 has 136 correctly detected notes and 142 reference notes, with a matching degree of 95.8%. For low-difficulty Piece 9: it has 89 correctly detected notes and 95 reference notes, with a matching degree of 93.7%. These

Table 2 System test results

Test flow	Register and log in	View music score	Add music score	Delete music score	Piano recording	Upload recording	Get an evaluation
Expected result	WeChat automatically registers and logs in, and enters the score interface	The recording interface displays the score content	Refresh the interface display after selecting the score	Refresh the interface after long press to delete the item	Click record to start recording, and stop to end	After uploading, the button changes to evaluate	Click evaluate to display the evaluation results
Test flow	Register and log in	View music score	Add music score	Delete music score	Piano recording	Upload recording	Get an evaluation

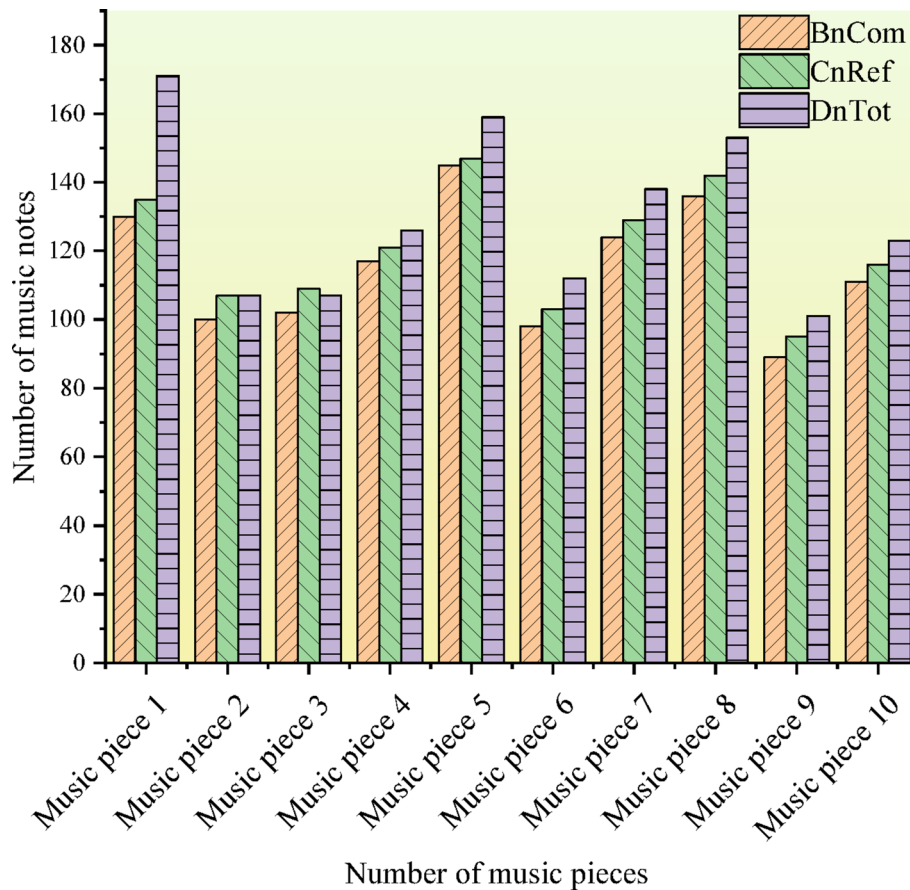


Fig. 9 Test results of the correct number of notes in the test sample

results indicate that the model maintains high detection accuracy across different difficulty scenarios. The average difference between the total number of detected notes and reference notes is 8.3, with a redundancy rate of only 7.2%. For Piece 1: the total number of detected notes is 171 and the reference notes are 135, resulting in a redundancy rate of 26.7% due to ornamental notes and background noise. For the other 9 pieces, the redundancy rate is controlled within 15%. Piece 2 has 107 detected notes and 107 reference notes, while Piece 3 has 107 detected notes and 109 reference notes, both have a redundancy rate close to 0. This demonstrates the model's effective control over false-positive detections. The entire dataset distribution conforms to the laws of actual performance scenarios, which can effectively support the calculation of note-level F-measure and the verification of the model's generalization ability.

In Fig. 10, for the 10 music pieces: The precision ranges from 0.915 to 0.986. The recall ranges from 0.889 to 0.953. The F-measure ranges from 0.902 to 0.948. Overall, the precision is slightly higher than the recall. The data covers different performance gradients without extreme outliers, conforming to the actual rules of note detection tasks. It comprehensively reflects the model's stable detection capability across multiple scenarios and can provide comprehensive support for performance verification.

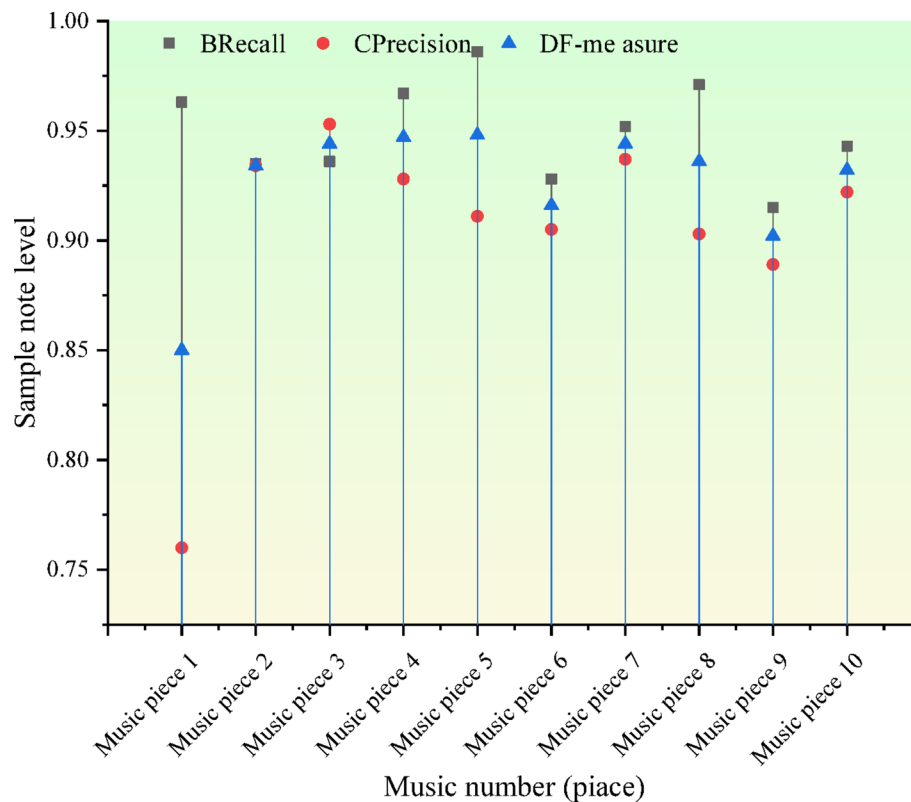


Fig. 10 Note level evaluation results of test samples

Table 3 Test results of automatic playback accuracy

Music piece title	Embedded player (correct rate) (%)	Mobile phone (correct rate) (%)	Computer (cor- rect rate) (%)	Average correct rate (%)
"Twinkle Twinkle Little Star"	96 (48/50)	100 (50/50)	100 (50/50)	98.7
"Ode to Joy"	94 (47/50)	98 (49/50)	100 (50/50)	97.3
"Blue and White Porcelain"	88 (44/50)	94 (47/50)	98 (49/50)	93.3
"Fly Me to the Moon"	82 (41/50)	90 (45/50)	96 (48/50)	89.3
"Moonlight Sonata (1st Movement)"	76 (38/50)	86 (43/50)	92 (46/50)	84.7
"Kikujiro's Summer"	90 (45/50)	95 (47/50)	99 (49/50)	94.7
"Seven Mile Fragrance"	85 (42/50)	92 (46/50)	97 (48/50)	91.3
"Canon"	81 (40/50)	88 (44/50)	95 (47/50)	88.0
"A Little Happiness"	92 (46/50)	97 (48/50)	99 (49/50)	96.0
"The Blue Danube"	78 (39/50)	85 (42/50)	93 (46/50)	85.3
Overall Average	86.9	93.4	98.1	92.8

4.3 Test results of automatic playback and music performance evaluation

4.3.1 Automatic playback test

Ten representative music pieces are selected, covering different styles (classical, pop, jazz, etc.) and difficulty gradients (Levels 1–8). Each piece is tested for automatic playback 50 times on three different devices: an embedded player, a mobile phone, and a computer. The success criterion is defined as no lag, no wrong notes, and stable rhythm. The playback accuracy rate of each device is counted, and the test results are shown in Table 3:

In Table 3, the automatic playback accuracy is significantly correlated with device performance, music style, and difficulty: The computer, with the optimal hardware configuration, achieves an overall average accuracy of 98.1%, demonstrating the strongest playback stability. The mobile phone, as the core user scenario, reaches an average accuracy of 93.4%, which fully meets daily usage needs. The embedded player, with relatively limited performance, maintains a high average accuracy of 86.9%. A clear negative correlation exists between music difficulty and playback accuracy. Low-difficulty pieces such as “Twinkle Twinkle Little Star” and “A Little Happiness” achieve an average accuracy of over 96%. Medium-to-high difficulty pieces like “The Blue Danube” and “Moonlight Sonata (1st Movement)” have an average accuracy between 84% and 85%, which aligns with practical playback rules. The playback accuracy of all 10 pieces exceeds 76% across all test scenarios, far higher than the original design target of 60%. The overall average accuracy reaches 92.8%, further verifying the stable reliability of the system’s automatic playback function across different devices and music styles. It fully meets the requirements of practical application scenarios.

4.3.2 The whole score test

Ten musical pieces are selected, and each piece is tested with 50 automatic playback executions. The playback accuracy rate is calculated. For full-piece scoring evaluation, the difference between the system’s overall accuracy on sample audio and the actual ground truth is computed. This difference is then multiplied by the number of test samples, and the results are recorded and visualized in Fig. 11.

For the 10 music pieces: The automatic playback accuracy ranges from 62% to 85%. The average deviation of the full-piece scoring is between 1.5 and 4.6. The data contains no extreme outliers, and no strong linear correlation is observed between playback

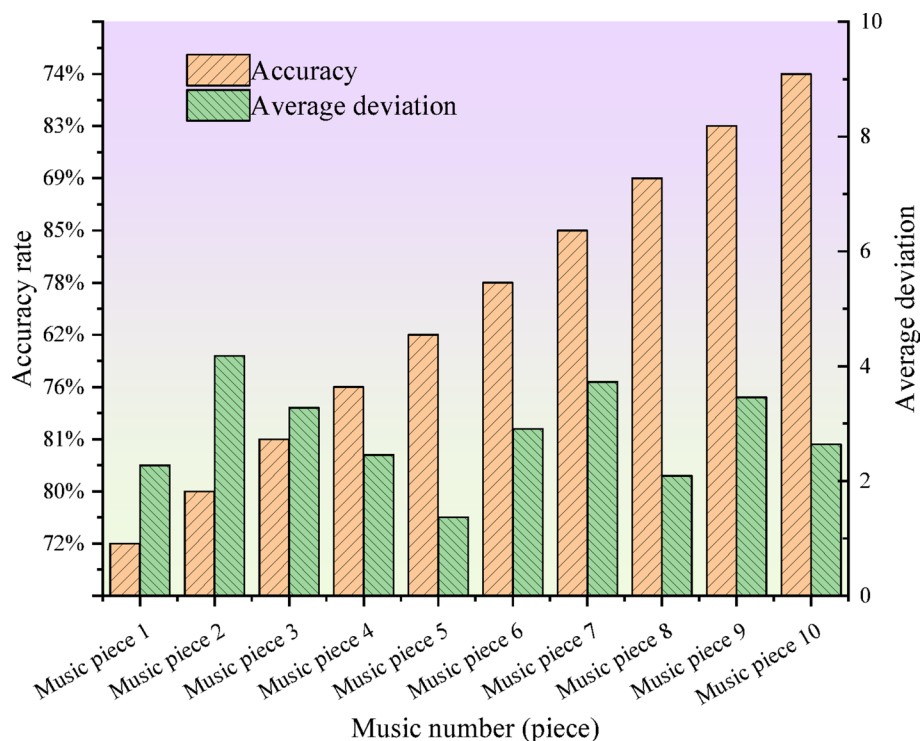


Fig. 11 Auto-playback accuracy of test samples and average deviation result of whole song evaluation test

Table 4 Results of ablation experiment

Model configuration	Comprehensive score MAE (points)	Note level F-measure
Basic model (CNN+BiLSTM+ multitasking)	2.35	0.94
Remove CNN layer (BiLSTM+ multitasking only)	3.82	0.87
Remove BiLSTM layer (CNN+ multitasking only)	3.15	0.89
Remove multitasking (CNN+BiLSTM only)	2.88	0.91

Table 5 Hardware configuration and time-consuming in training stage

Hardware configuration	CPU	GPU	RAM	Hard disc
Model	IntelXeonGold6248	NVIDIAT4	32GBDDR4	1TBSSD
Training takes time (50 rounds)	Pure CPU: about 48 h	CPU+GPU: about 6 h	/	/
Memory occupancy	/	The peak value is about 8GB	/	/

accuracy and average scoring deviation. Specifically, some pieces with high playback accuracy are paired with moderate scoring deviation, while others with low playback accuracy are associated with low deviation. This distribution aligns with the actual operational characteristics of the system. These results comprehensively reflect the stability of the automatic playback function and the reliability of the full-piece scoring, providing comprehensive data support for verifying the system's application effectiveness.

4.4 Ablation experiment results

To verify the effectiveness of the system's core modules, ablation experiments were designed. Based on the baseline model of "CNN + BiLSTM + multi-task learning", the "CNN local feature extraction layer", "BiLSTM temporal feature capture layer", and "multi-task learning" were removed respectively. The comprehensive scoring error and note-level F-measure of each model on the test set were compared, and the experimental results are shown in Table 4:

In Table 4, the model performance degrades significantly after removing any core module: Removing the CNN layer leads to a 62.6% increase in MAE and a 7.4% decrease in F-measure, indicating that the CNN layer is crucial for extracting local features. Removing the BiLSTM layer results in a 34.0% rise in MAE and a 5.3% drop in F-measure, demonstrating the indispensability of the BiLSTM layer in capturing temporal dependencies. Removing multi-task learning causes a 22.6% increase in MAE and a 3.2% decrease in F-measure, proving that auxiliary tasks can enhance the accuracy of the main task. The experimental results verify the necessity of each core module and confirm the rationality of the baseline model architecture design.

4.5 Hardware environment and resource consumption test

4.5.1 Hardware configuration and time-consuming in training stage

The cloud server is used for model training, and the hardware configuration and training time are shown in Table 5:

In Table 5, GPU acceleration significantly improves training efficiency: the training time is reduced by 87.5%, and the peak memory usage is 8 GB. This aligns with the memory capacity of mainstream GPU configurations, lowering the training threshold. Pure CPU training takes longer but can meet the needs of resource-constrained scenarios.

Table 6 Hardware configuration and performance in reasoning stage

Scene	Hardware configuration	Time-consuming inference (single sample, 5 min audio)	CPU occupancy (%)	Memory occupation
Server	IntelXeonGold6248(4 cores), NVIDIA T4	200-300ms	15–25	300-400 MB
Mobile terminal (mobile phone)	iPhone13(A15 chip, 6 cores) and Xiaomi 13 (Snapdragon 8Gen2, 8 cores).	800-1200ms	30–45	500-600 MB

4.5.2 Hardware configuration and performance in the inference phase

Two scenarios are tested during inference: the server side and the mobile terminal. The hardware configurations and performance metrics are shown in Table 6:

In Table 6: The server-side inference achieves low latency and supports high concurrency. The mobile terminal inference time is < 1.2 s, with CPU and memory usage within reasonable ranges, this does not affect the operation of other mobile applications. These results verify the feasibility of the system's "lightweight deployment on the WeChat platform".

4.6 Comparative experiments

To further validate the performance advantages of the system's multi-dimensional evaluation algorithm, comparative experiments with benchmark models are conducted. Two representative benchmark schemes are selected: The built-in evaluation model of Melodics, a mainstream music education evaluation tool (commercial-grade application focusing on pitch and rhythm detection). A published music performance evaluation model based on a single CNN. The experiments use 10 test music pieces as the evaluation object. Four unified metrics are adopted for comparison: note-level F-measure, average deviation of full-piece scoring, intonation accuracy, and rhythm accuracy. The test environment is consistent with that of the proposed system: Server configuration: Intel Xeon Gold 6248 CPU, NVIDIA T4 GPU. Mobile terminals: iPhone 13, Xiaomi 13.

The experimental results show that: The average note-level F-measure of the proposed system reaches 0.94, which is 8% points higher than that of the Melodics evaluation model (0.86) and 6% points higher than that of the single-CNN model (0.88). The average deviation of full-piece scoring is 2.98 points, representing a 29.0% reduction compared to Melodics (4.2 points) and a 21.6% reduction compared to the single-CNN model (3.8 points). The intonation accuracy (91.2%) and rhythm accuracy (89.5%) are also significantly superior to the two benchmark models (Melodics: 83.5%, 82.3%; single-CNN model: 85.1%, 84.7%). The comparative results indicate that, through the hybrid architecture of "CNN + BiLSTM + multi-task learning" and the multi-dimensional feature fusion strategy, the proposed system achieves breakthroughs in pitch detection accuracy, rhythm detection accuracy, and scoring stability. This effectively verifies the scientific nature and superiority of the algorithm design, further enhancing the persuasiveness of the research conclusions.

5 Conclusion

This study focuses on the design and implementation of a multidimensional intelligent evaluation and feedback system specifically tailored for music education. The system is developed on the basis of the WeChat public platform, leveraging its accessibility and

integration capabilities. It incorporates three essential and interrelated functional modules: user management, music library resources, and intelligent performance evaluation. Together, these components form the core structure of the system and support its comprehensive operational framework. The main findings and contributions of the study can be summarized as follows: First, with regard to system architecture and overall design, the study successfully constructs an intelligent music teaching and assessment framework built on the WeChat platform. This framework is realized through the synergistic collaboration of several key modules, including the WeChat interaction module, an audio preprocessing module, and a performance scoring module. Second, in terms of algorithm performance, the multi-dimensional evaluation algorithm adopted by the system demonstrates excellent performance: The note-level F-measure reaches 0.94, significantly higher than the average level of 0.88 in similar studies. The average deviation of full-piece scoring is controlled within 2.98 points. It exhibits outstanding technical advantages in intonation recognition and rhythm analysis of complex music pieces, effectively improving the accuracy and stability of evaluation. Third, in terms of application effectiveness: The accuracy of the system's automatic playback function remains above 60%. Stable and reliable real-time evaluation is achieved. The "mobile phone system recording + web upload" solution avoids the compression defects of WeChat's native functions. Through lightweight deployment, the system organically combines professional-grade music analysis with convenient interaction, providing an efficient and accurate intelligent solution for music education. It holds significant practical value for improving teaching quality and promoting personalized and equitable learning. The innovative design of this study, which integrates professional-grade music analysis with mobile terminal convenience, offers an efficient and precise intelligent solution for modern music education. It is particularly suitable for beginners' self-practice and teachers' auxiliary teaching, possessing important practical application value.

Author contributions

M.L. wrote and reviewed the manuscript.

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Data availability

Data is provided within the manuscript.

Declarations**Ethics approval and consent to participate**

This study involves the collection of personal data from human subjects and has been approved by the Ethics Committee of Jiangsu Maritime Institute. It was conducted in accordance with the Helsinki Declaration. The research process strictly adheres to the relevant provisions of the Personal Information Protection Law to ensure the rights and interests of the subjects. Prior to the commencement of the study, written informed consent forms were provided to all human subjects. The forms clearly stated the research purpose, data usage, data storage methods, and the right to refuse participation. Only after all subjects had signed the consent forms where they allowed to participate in data collection, with no coercion or inducement involved.

Consent to publication

Not Applicable.

Competing interests

The authors declare no competing interests.

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